



ROBOT EMBEDDED PROGRAMING

Lab 1: Review of Linux and Python for Robotics Applications



Name:

Student #1

Student #2

Student #3

Student #4

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Objective

Introduce students to Linux, Python, and their importance in robotics, with a focus on building shared knowledge before starting technical work in later labs.

Session Outline

1. Welcome & Course Context (10 min)

- Overview of how the lab series fits into the course.
- Explanation of how Linux and Python connect to ROS 2 and real-world robotics.
- Short video or slides showing ROS 2 in action.

2. Linux in Robotics – Discussion (15 min)

- Instructor-led questions:
 - Where is Linux used in industry?
 - Why do many robotics systems use Linux instead of other operating systems?
- Show simple examples of Linux interfaces (terminal and GUI) on the projector.

3. Python for Robotics – Discussion (15 min)

- Talk about Python's popularity and why it's chosen for ROS 2 development.
- Mention common robotics-related libraries (e.g., rclpy, OpenCV, NumPy).
- Example: Python controlling a simulated robot (shown by instructor, not run by students).

4. Group Activity: Research & Whiteboard Sharing (30 min)

- Step 1: Students form small groups of 3–4.
- Step 2: Assign each group one of these topics:
 1. Advantages of Linux for robotics.
 2. Python's role in robotics.
 3. Key differences between ROS 1 and ROS 2.
 4. Examples of industries and projects using ROS 2.
- Step 3: Groups spend 15 minutes researching online or discussing based on prior knowledge.
- Step 4: Each group writes their 3–5 key points on the whiteboard.
- Step 5: Class discussion to compare findings.

5. Introduction to ROS 2 Jazzy Documentation (10 min)

- Navigate together to the official ROS 2 Jazzy docs:
<https://docs.ros.org/en/jazzy/index.html>
- Show where to find:
 - Tutorials
 - Concepts (Nodes, Topics, Services, Parameters)
 - Example projects
- Explain that no installation is needed yet—this will be done in Lab 2.

Deliverables

- Participation in group activity and class discussion.
- Students should leave with personal notes on:
 - Linux's importance in robotics.
 - Python's role and benefits.
 - One interesting fact learned from another group.

Preparation for Lab 2

- Review the "Beginner: CLI Tools" section of the ROS 2 Jazzy docs.
- Bring your laptop ready for installation activities next week.